

quickly take over the bulk of network traffic slowing down everything else. If things get out of control, this could lead to file sharing being banned on your LAN. To prevent that it would be a good idea:

1. not to use the server or supernode mode
2. to limit downloads and uploads to one file at a time
3. to schedule downloads to off-peak hours
4. not to use P2P clients with adware as they can generate additional traffic even when nothing is being downloaded or uploaded

13.2 BitTorrent

Unlike other file sharing protocols, BitTorrent does not require the entire file to begin sharing the file. BitTorrent splits a file into smaller fragments which are uploaded to different peers (users). The protocol is smart enough to request the rarest fragment first from peers with the best network connections. This increases the overall efficiency of the “swarm” (a group of peers sharing a full or partial BitTorrent file) and makes for faster downloads and propagation of the complete file to as many peers as possible in the shortest possible time.

Also, unlike other file sharing protocols where you simply drag a file into a shared folder, BitTorrent requires some additional steps to create a “torrentable” file. There are four elements involved in making your file available via BitTorrent:

- a) A .torrent file. This is a metadata file containing the names, sizes, and checksums of all the pieces that make up the shared file or directory.
- b) A seeder. This is a peer who has a complete copy of the actual file being shared. When you first make your file available via BitTorrent you will be the seeder.
- c) A tracker. This is a server that coordinates connections among the seeders and peers. Clients report to the tracker, which in turn informs them of other available peers. Note that you can